

YOUSSEF HASSAN

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Summary

Backend Software Engineer with hands-on experience developing scalable backend systems, RESTful APIs, and AI-powered applications using ASP.NET Core, FastAPI, Python, and SQL Server. Experienced in integrating Computer Vision and Large Language Models into real-world systems, with a strong foundation in software architecture, problem-solving, and competitive programming (ECPC).

Education

2022 – 2026

Bachelor's in Computer Science and Artificial Intelligence

Cairo University, Giza, Egypt | 2022 – 2026

Graduation Project: Basira AI – Assistive System for Visually Impaired Children (Excellent)

Experience

Jun 2026 – present

Backend / AI Engineer (Project-Based) Global Auto (BMW)

- Architected and developed an enterprise-grade AI-powered Text-to-SQL data warehouse assistant, enabling non-technical employees to query complex databases using natural language (Arabic and English).
- Designed a secure RESTful API backend using Python and FastAPI, integrating local Large Language Models (LLMs) via Ollama with a custom deterministic NLP pipeline to ensure 100% SQL safety and eliminate AI hallucinations.
- Implemented Role-Based Access Control (RBAC) to enforce data privacy, restricting sensitive database access and query execution based on employee authorization levels.
- Optimized database interactions with Microsoft SQL Server by engineering a semantic resolution layer and query planner, significantly improving data accessibility and reporting efficiency for business teams.
- Deployed and maintained the full-stack application on enterprise Virtual Machines (VMs) via Remote Desktop (RDP), ensuring system stability, seamless local network integration, and optimized resource utilization.

Apr 2024 – Mar 2026

Prompt Engineer at Scale-AI Company

- Designed and optimized complex AI prompts applying Context Engineering and Harness Engineering techniques to improve model reasoning and output quality.
- Collaborated with teams to refine system responses and enhance accuracy
- Worked under pressure to meet performance and quality targets

Competitions & Certifications

2023. 2024

Egyptian Collegiate Programming Contest (ECPC) Qualifications

- Solved algorithmic problems under time constraints and strengthened problem-solving.

Projects

Basira AI – Assistive System for Visually Impaired Children

(Python | FastAPI | YOLOv8 | MiDaS | FaceNet | BLIP | Florence-2 | Flutter | ESP32-CAM)

- Developed an end-to-end AI-powered wearable platform integrating FastAPI backend, REST APIs, WebSockets, Flutter applications, and ESP32-CAM for real-time indoor navigation and bilingual voice guidance.
- Built a production-ready computer vision pipeline using YOLOv8 through dataset engineering, annotation, augmentation, class balancing, transfer learning, fine-tuning, and model evaluation for indoor object detection.
- Implemented depth estimation, scene understanding, and face recognition using MiDaS, BLIP, Florence-2, and FaceNet to provide contextual scene summaries, distance estimation, obstacle awareness, and familiar person identification.
- Engineered intelligent Safety & Emergency Alert workflows supporting dangerous object detection, proximity-based alerts, parent monitoring, QR-based device pairing, and real-time notifications.
- Designed an integrated AI pipeline enabling real-time inference, AI model orchestration, hardware-backend communication, and end-to-end system testing.

Enterprise AI Data Assistant (Python | FastAPI | SQL Server | React, LLMs)

- Built a full-stack internal tool to translate conversational queries into executable SQL statements against a live Data Warehouse, featuring automated data visualization (Matplotlib/Seaborn) and dynamic React/TypeScript dashboards.
- Engineered a multi-stage semantic pipeline using Advanced Regex, Intent Classification, and LLM Prompt Engineering to accurately map unstructured business terms to relational database schemas.
- Enforced strict system security by developing a custom SQL AST (Abstract Syntax Tree) Validator to block destructive commands (e.g., DROP, DELETE) and ensure read-only query execution.

E-Commerce System API (.NET | ASP.NET Core | EF Core | SignalR | xUnit)

- Developed an enterprise-grade RESTful API following Clean Architecture principles. Implemented secure Role-Based Access Control (RBAC), real-time order notifications using SignalR, and robust data access via Repository and Unit of Work patterns, while ensuring system reliability through comprehensive unit testing.

Inventory Management System (.NET, EF Core)

- Developed a backend inventory management system using EF Core, implementing CRUD operations, filtering, sorting, pagination, Repository Pattern, Unit of Work, and relational database design.

School Management System (.NET, EF Core)

- Developed a backend school management system using ASP.NET Core and EF Core, implementing CRUD operations, validation, business rules, modular architecture, Controllers, DbContext, and filtering.

Movie Tracker SPA (Laravel | MVC | MySQL | PHPUnit | OMDb API)

- Developed a full-stack Laravel MVC application integrating Eloquent ORM, Blade, OMDb API, and PHPUnit for movie management, real-time data retrieval, and automated testing.

Embedded Systems Projects (ATmega32)

- Designed and implemented embedded systems including Door Locker Security System, Distance Measuring System, Temperature-Based Fan Controller, and Digital Stopwatch using ATmega32, implementing UART, I2C, EEPROM, GPIO, Timers, Interrupts, PWM, Motor Control, and interfacing with HC-SR04 Ultrasonic Sensor, LM35 Temperature Sensor, PIR Sensor, and Buzzer.

Courses

Apr 2026

Backend Development Diploma (.NET)

- Built backend systems using ASP.NET Core and Entity Framework Core, implementing RESTful APIs and CRUD operations, applying the Repository Pattern and Unit of Work, and working with SQL databases, validation, filtering, caching, authentication, and SignalR.

Aug 2024

Full Embedded Systems Diploma under the supervision of Engineer Mohamed Tarek

- Developed embedded projects using AVR (ATmega32), implementing low-level drivers and real-time applications, with hands-on experience in RTOS and embedded systems design.

Technical Skills

Programming Languages	C/C++, Python	C#, PHP, TypeScript
Backend and APIs	- ASP.NET Core - REST APIs - FastAPI - Authentication & Authorization - WebSockets - Unit of Work	- Entity Framework Core - Web Development (HTML, CSS, JavaScript) - JWT - Role-Based Access Control (RBAC) - Clean Architecture - Repository Pattern - SignalR
Databases	- SQL Server (SSMS) - SQLite	MySQL
AI & Data Engineering	- Computer Vision - Large Language Models (LLMs) - Text-to-SQL - Data Warehousing - Real-Time Inference - Model Fine-Tuning	- Prompt Engineering - Context Engineering - NLP - Transfer Learning - NL2SQL
Embedded Systems	- Embedded C - Interfacing	- AVR (ATmega32) - Communication Protocols (UART, SPI, I2C)
Core Concepts	- Data Structures and Algorithms - SOLID Principles - Problem Solving	- Object Oriented Programming (OOP) - Software Engineering - System Design
Tools	Visual Studio / Visual Studio Code	Git/GitHub